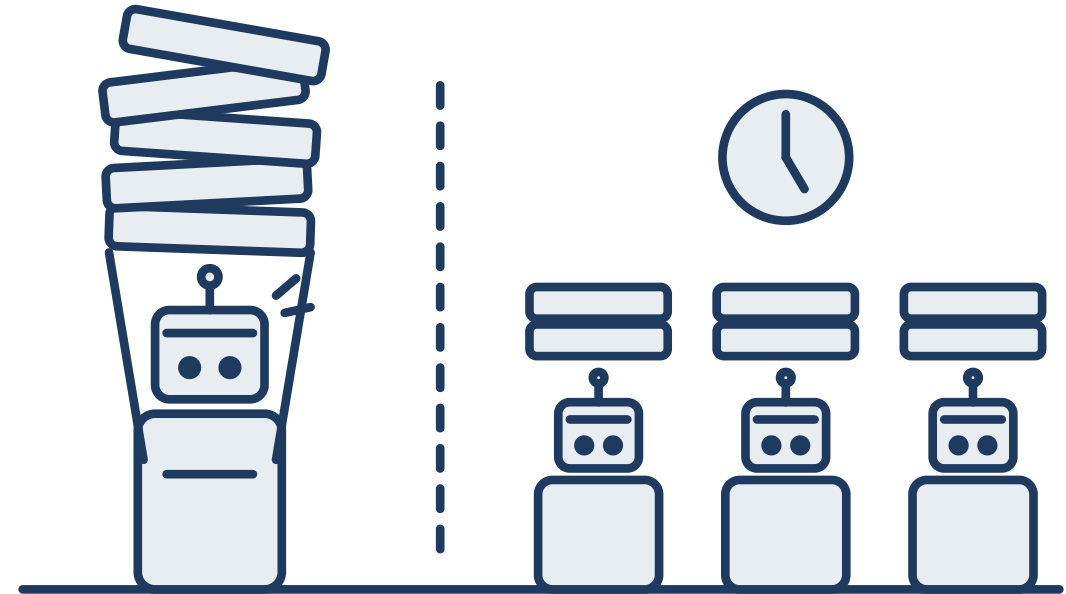


Multi-agent workflows

Shoubaneh Hemmati, Caltech/IPAC.

GRITS AI workshop, September 2026.

Repository: github.com/xoubish/multiGrits



One collaborator, then more

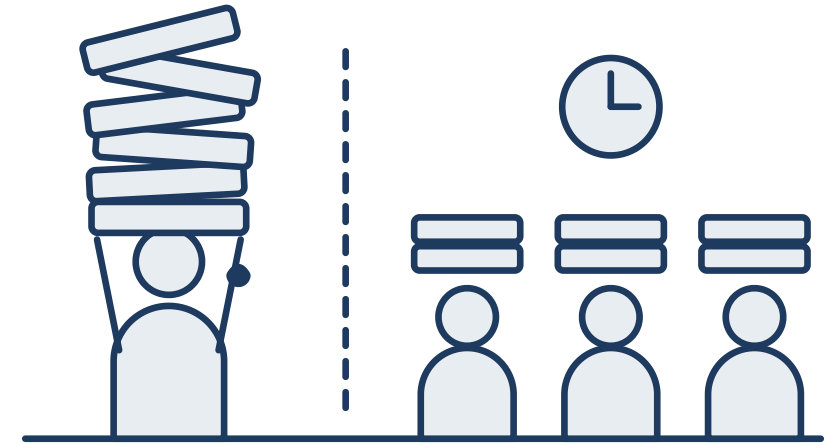
This talk assumes prior use of a coding agent such as Claude Code or Codex on real work. A single agent already plans, writes, checks, and debugs on request, and it has been the best collaborator I have had.

When one collaborator works that well, the obvious next step is a team of them. This talk poses that question before answering it.

Why do people work in teams?

People work in teams for many reasons, some of them social. Four bear on agents, and cost shapes all four.

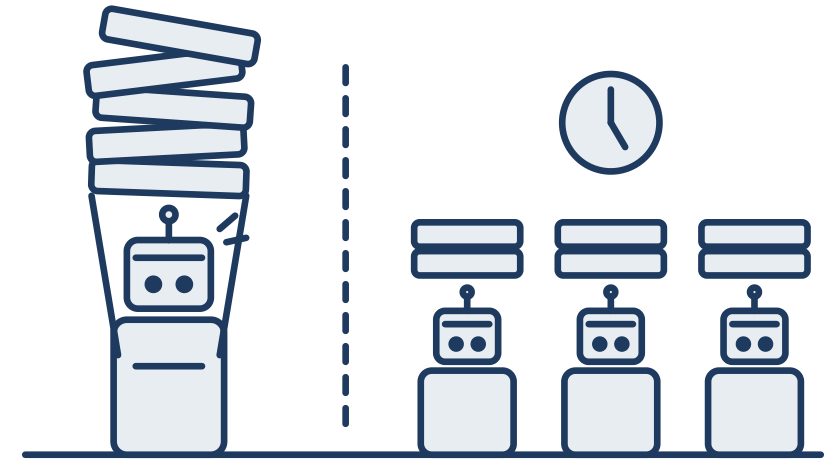
- Time is limited, so the work is divided.
- Knowledge is limited, so expertise is brought in.
- Attention is limited, so pieces are handed off (Simon 1971).
- Judgment is anchored to our own drafts, so others check them.
- Cost sets the mix: one Einstein might do the work of five.



Which of those reasons hold for agents?

Three of the four hold for agents; one holds in part.

- Time holds: agents run in parallel.
- Knowledge holds in part: copies of one model know the same things; other models or tools add some.
- Attention holds: each agent brings a fresh context window.
- Judgment holds: a second agent is not anchored to the first draft.
- Cost sets the mix here too: several cheap models for the price of one frontier model, which often wins anyway.



What multi-agent brings: more context than intelligence

Multi-agent buys more context than intelligence. Windows have grown about five hundredfold in six years; the usable window has not kept pace, and most tasks need none of this.

Year	Model	Context window	Source
2020	GPT-3	2,048 tokens	Brown et al. 2020
2024	GPT-4 Turbo; Claude 3	128K; 200K	Gemini Team 2024
2024	Gemini 1.5	retrieval tested to 10M, in research	Gemini Team 2024
2026	Claude Fable 5.1, Opus 5, Sonnet 5	1M	Anthropic 2026, models page

In English prose a token is about 0.55 to 0.75 of a word, so 1M tokens is roughly 550,000 to 750,000 words (Anthropic 2026).

What fills a context window

- Each turn re-sends the whole window: system prompt, tool definitions, messages both ways, and every tool result.
- Tool results dominate: 580K of the 613K tokens in use in this deck's build session.
- When it fills, Claude Code compacts: the older transcript becomes a summary, and unsaved detail is lost.
- A subagent's transcript stays in its own window; only its return enters the parent's.

Context usage

claude-fable-5-1 · 613.1k / 1.0M tokens (61%)



Category	Tokens	Share
System prompt	4.9k	0.5%
System tools	23.3k	2.3%
Agents, memory, skills	5.2k	0.5%
Messages and tool results	580.4k	58.0%
Autocompact buffer	33.0k	3.3%
Free space	353.3k	35.3%

Claude Code /context panel, this deck's build session, 2026-09-18. Redrawn; figures as shown.

The usable window is smaller than the advertised one

- Accuracy fell more than twenty points when the answer sat in the middle of the context, below the score with no documents at all (Liu et al. 2023).
- GPT-4 advertised 128K tokens; its effective length was about 64K (Hsieh et al. 2024).
- Models effectively use only 10 to 20 percent of their context (Kuratov et al. 2024).
- Without literal word overlap between question and answer, 11 of 13 long-context models fell below half their short-context accuracy by 32K tokens (Modarressi et al. 2025).
- Human memory shows the same position curve, but the similarity is shallow: people compact to gist within seconds and keep almost no wording, while a model keeps every token and loses the ability to attend to them (Jarvella 1971; Guo and Vosoughi 2025).

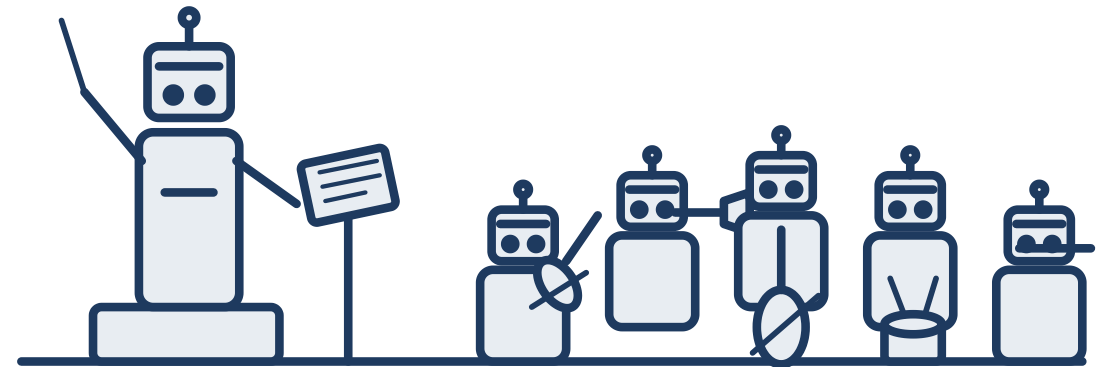
One strong model with a long window

- Given enough budget, one long-context model consistently outperforms retrieval over pieces of the material, at higher cost (Li et al. 2024).
- At an equal thinking budget, a single agent matched or beat five multi-agent designs (Tran and Kiela 2026).
- Splitting pays only when the work exceeds what one strong window handles well, or when independence or wall-clock matter.

Elements of a multi-agent system

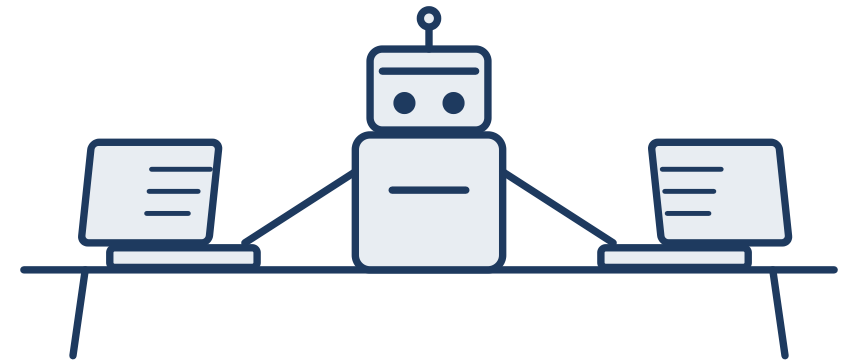
Designing a multi-agent system means thinking along three axes, taken one at a time on the next three slides.

- **Who orchestrates:** a script, a model, or a human.
- **Topology:** how the agents are connected.
- **Isolation:** shared or fresh context; shared files or separate git worktrees.



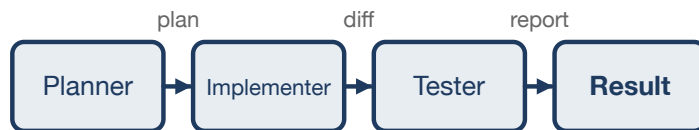
Axis 1, who orchestrates: a script, a model, or a human

- A script fixes the sequence in advance: which agent runs, in what order, how many rounds. Anthropic calls this a workflow.
- The model chooses each next step at run time, including whether to call another agent. Anthropic calls this an agent (Anthropic 2024).
- A person can orchestrate too: two terminals, the human as merge point.
- Scripts are reproducible; a model deciding is flexible but hard to reproduce.

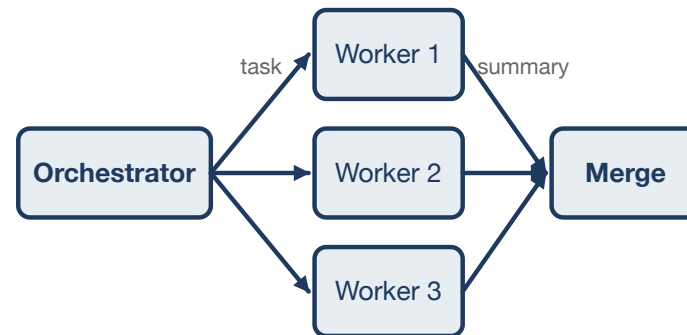


Axis 2, topology: how the agents are connected

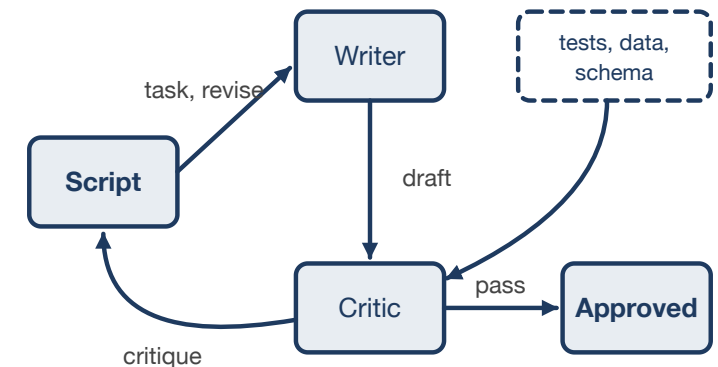
Three shapes cover most systems. Errors are caught at the merge point, or not at all (Kim et al. 2026).



Chain: each stage hands its output to the next, files as the hand-off. A planner writes the plan, an implementer the code, a tester runs it.



Fan-out and merge: independent pieces, one merge point. Each worker returns a summary, not its raw material.



Loop: writer and critic pass work back and forth a fixed number of times. The critic holds the tests and data; a critic that can only read prose agrees with it.

Axis 3, isolation: what the agents share

- Context: shared, so one agent carries everything; or fresh, so each stage starts empty and receives only what it needs. Fresh context is the gain the previous section argued for.
- Files: shared, so two agents can write the same file; or one git worktree per agent, merged at the end.
- A Claude Code subagent starts with a fresh context and receives only the delegation message (Anthropic 2026).
- The failure isolation prevents is two agents writing one file; an instance appears in attempt one.

The more the merrier?

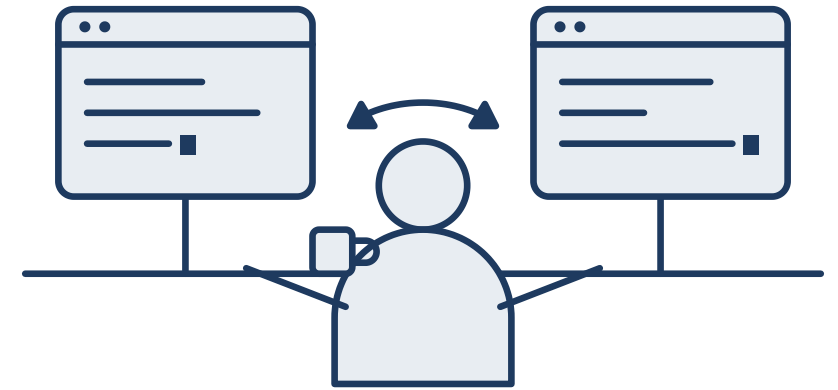
Scale	What it is	Who
1	The default. One conversation; a subagent is spawned only when the model decides to (Anthropic 2026).	Everyday use
2 to 7	One orchestrator and a few workers: a lead plus subagents (Anthropic 2025), five fixed roles (Hong et al. 2024), seven (Qian et al. 2024); three to five teammates is the vendor's own guidance (Anthropic 2026).	Working systems
Tens	Sampling and voting keeps improving into the tens of samples, but 40 samples of a weak model still trail one call of a strong one (Li et al. 2024), and returns diminish once the single agent is strong (Kim et al. 2026).	Benchmarks
1,000+	Collaboration among over a thousand agents (Qian et al. 2024); simulations of 10 to 1,000+ (Altera 2024). Nobody uses this for work.	Research

There is no published survey of what practitioners run; the "Who" column is an assessment.

You already do this

- The same task in Claude Code and in Codex, then a comparison: writer and critic, you as orchestrator.
- Two terminals on two tasks, then a merge by hand: fan-out and merge.
- A new session for the next phase, handed a plan file: a chain, fresh context.
- Two clones of the repository for two experiments: isolated workers.
- Claude Code spawning an Explore subagent unasked: a fan-out nobody chose.

The question: when to make it deliberate, and when to hand the orchestration to a script.



This deck was built by the pipeline in this repo

Every part of this deck, the slides, the diagrams, the evidence, the notes, and the Q&A, was drafted by ten agents in this repository, called in a fixed order by a shell script, with every call logged. What follows is the recipe, beginning with what went wrong.



The repository

This is the repository layout from the README, trimmed to the files that carry the story. The full listing is in the handout and in the repository, github.com/xoubish/multiGrits.

<code>talk-context.md</code>	audience, schedule, non-goals, thesis, style rules for every agent
<code>slides/outline.md</code>	HUMAN-WRITTEN. Slide order, per-slide message, time budgets, cuts. No agent edits it.
<code>.claude/agents/</code>	ten subagent definitions (see table below)
<code>pipeline/run.sh</code>	deterministic orchestration: stages, the review loop, worktree isolation
<code>research/evidence.md</code>	one source per claim in the outline (written by evidence-finder)
<code>research/build-log.md</code>	how this deck was built, step by step, with pointers into runs/ (written by chronicler)
<code>diagrams/</code>	Mermaid, one file per diagram token in the deck
<code>runs/</code>	one directory per agent call: exact prompt, full JSON result, return text, cost row

Attempt one, from scratch

The first pipeline let agents write the outline, not just the slides. Eleven agent files did the work: four researchers in parallel, an outliner, a slide-writer and a diagrammer in worktrees, three critics for content, design, and teaching, a fact-checker, a notes-writer, a Q&A skeptic, and a demo editor.

Runs 002 through 014 cost \$29.22 over 455 turns.

Attempt one, failure 1: the spec contradiction

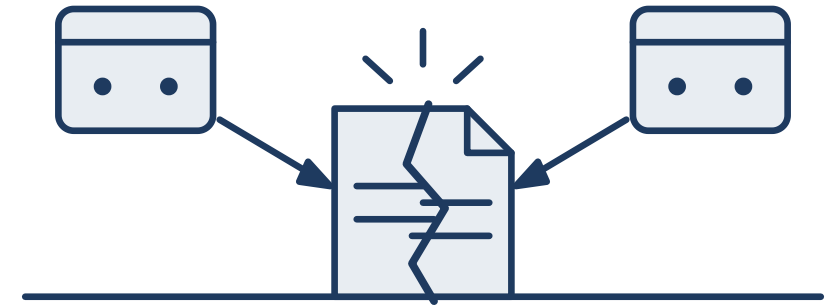
In run 002, the outliner found that my segment table summed to 27 minutes of content plus 3 of Q&A, my prose said 25 plus 5, and its own instructions demanded 1500 seconds. It could not satisfy all three, so it picked one, scaled three segments down, and logged the choice in its notes rather than choosing silently.

The fix was to the spec, and the stage was re-run as run 003.

Attempt one, failure 2: the shared cost log

In run 004, the slide-writer and the diagrammer ran in two worktrees and never touched each other's outputs. But `log_result.py` appended one row to a single shared `runs/cost.tsv` from inside each worktree, so the merge conflicted on the pipeline's own bookkeeping.

The fix was for each run directory to write its own `cost-row.tsv`, with a report regenerated from every `result.json`.



Attempt one, failure 3: the budget cap

In run 006, the first revise pass exhausted the then-default \$3 `--max-budget-usd` ceiling after 34 turns. `result.json` said `terminal_reason: budget_exhausted`, but the exit code was 1, not the 2 the docs describe. It had already applied 20 of its 24 changes.

The fix was to raise the default budget to \$5 and have the script read `result.json` instead of trusting the exit code.

What came out

Every constraint was met: citations complete, word counts under the cap, segment timing to the second. The fact-checker confirmed 44 citations, 2 partial, 0 not found.

And it was a wall of cited percentages, plus a twelve-minute demo of screenshots of README files. This is one of its slides.

Gotcha · Isolate the files, centralize the merge

Two agents editing one file is the classic failure. Worktrees fix it [Claude Code worktrees].

Independent agents with no central check amplified errors 17.2x; with centralized verification, 4.4x [Kim 2026].

"The bottleneck is no longer generation, it's verification" [Osmani].

What came out, and why

The three critics scored what could be counted: citation counts, words per second, and seconds per slide. Nobody scored whether it was a talk.

Agents optimize the rubric they are given. Taste is not in the rubric, and it cannot be, so taste has to be a human's.

Why multi-agent: context, not intelligence

- Answer placed mid-context: accuracy drops over 20 points, below the no-documents score [Liu 2023]
- GPT-4 claimed 128K tokens; effective 64K on a long-context test [Hsieh 2024]
- Fix: delegate bounded subtasks; each subagent returns a 1,000–2,000-token summary [Anthropic 2025a]

The reset

The pipeline was reset. I wrote this outline by hand, and no agent generates or reorders it. This is its first entry exactly as it sits in `slides/outline.md` ; every entry has that shape:

```
1. Title Multi-agent workflows. Shoubaneh Hemmati, Caltech/IPAC. GRITS AI workshop, September 2026. 15 s.
```

Agents draft slides, diagrams, evidence, notes, and reviews for that outline. If one finds a structural problem, it reports it to me under a "For the speaker" heading and does not fix it.

The reset: retired and added

Retired were the outliner, because I own the outline; the four researchers, whose push-mode briefs nobody used; the three critics, three rubrics that rewarded compliance; and the demo editor.

Added were an evidence-finder that sources only the claims I make, an example-builder that runs the take-home example for real, one reviewer that sits in the audience, a chronicler that records the build, and an illustrator.

Recipe, part one: an agent is a markdown file

This is `.claude/agents/reviewer.md`, trimmed to fit. The frontmatter names the agent, says when to use it, lists its tools, and picks the model. The description is cut short here with an ellipsis; the full line, and the other nine files, are in the repo and in the handout.

```
---  
name: reviewer  
description: Sits in the audience. Reads the deck as a skeptical IPAC engineer  
             and looks at the rendered slide images. [...]  
tools: Read, Glob, Grep, Write  
model: inherit  
---
```

Recipe, part one: the instructions

Below the frontmatter come instructions in plain English, like a brief to a colleague. The last line matters, because the script parses it to decide whether to loop.

```
You are the one reviewer. You replace three earlier critics whose rubrics counted citations, words per second, and seconds per slide; the deck they passed was unpresentable. Do not count things. Judge whether the talk works.
```

```
5. Is there a moment where the speaker admits something did not work? If not, say where one belongs.
```

```
Write to the run directory you are given as `review.md`. First line exactly `Verdict: PASS` or `Verdict: REVISE`.
```

Recipe, part two: a script calls it headless with a budget

This is the call from `pipeline/run.sh`, exactly. `--agent` picks the markdown file, `--output-format json` returns a parseable result with the cost in it, `--max-budget-usd` caps the spend, and `--no-session-persistence` means no state carries over between calls.

```
env -u CLAUDECODE claude -p --agent "$agent" \  
  --output-format json \  
  --permission-mode acceptEdits \  
  --allowedTools "$tools" \  
  --max-budget-usd "$BUDGET" \  
  --no-session-persistence \  
  "$(cat "$run_dir/prompt.md")" > "$run_dir/result.json"
```

Recipe, part two: the stage order

```
evidence → example → chronicle → write (slide-writer + diagrammer + illustrator, 3 worktrees, merged)
→ loop (critique → revise, until PASS or 2 rounds) → chronicle → revise → factcheck
→ notes & qa (parallel) → cost
```

Evidence, example, and chronicle run one after another, which is a pipeline. Write is pattern 4, three worktrees merged. The loop is writer and critic, and the script owns the loop. Chronicle runs again, and one revise pass places the new build-log entries. Notes and Q&A are fan-out. Cost is a Python script and uses no model at all.

Attempt two: runs 015 and 016

Run 015, evidence. `evidence-finder`, Haiku 4.5 and Sonnet 5, read the outline and the old briefs and wrote `research/evidence.md`. It sourced 13 of 16 claims from existing material and flagged 4 as unsupported rather than guessing. 37 turns, \$0.7286, 171 seconds.

Run 016, example. `example-builder`, the only agent with Bash, built and ran the exoplanet lookup shown in the simple example. 28 turns, \$0.3209, 207 seconds.

Attempt two: runs 017 and 018

Run 017, chronicle. `chronicler`, Haiku 4.5 and Sonnet 5, read every run directory, the cost report, the agent files, the outline, and the README, and wrote the first `research/build-log.md`. 38 turns, \$0.4361, 128 seconds.

Run 018, write. A fan-out with isolated workers: `slide-writer`, `diagrammer`, and `illustrator` in three git worktrees at once, merged by the script with no conflict. 42 turns, \$4.1925 for the stage.

Attempt two: runs 019 and 020

Run 019, critique. reviewer, session model, read the deck and all 54 rendered slide images. 64 turns, \$2.9727, 150 seconds. Verdict: REVISE. The illustrations were never placed, two code blocks rendered at about 7 pixels, and two diagrams were squeezed unreadable.

Run 020, revise. slide-writer placed the illustrations, fixed the code blocks, and took the diagrams full width. 27 turns, \$1.6625, 132 seconds.

Attempt two: runs 021 and 022

Run 021, critique. reviewer again: 63 turns, \$2.9859, 171 seconds. Verdict: REVISE. Two illustrations were cover-cropped, the Tran and Kiela citation still lacked a title, and one sentence about diminishing returns had no source.

Run 022, revise. slide-writer fixed the crops, split a fourteen-line code slide in two, and rewrote that sentence as my own inference. 29 turns, \$1.4847, 100 seconds.

Attempt two: runs 023 to 029

Run 023, chronicle. The second chronicler pass wrote the build log for runs 015 through 022. 77 turns, \$0.77.

Runs 024 to 027. One revise pass placed those entries; the fact-checker verified every source in the deck; the notes-writer and the Q&A skeptic wrote the handout. Together 112 turns, \$4.20.

Run 029, revise. A final editorial review, written by the speaker rather than the reviewer agent, made the posted slides stand alone. 12 turns, \$3.84.

What it cost, and the receipts

Attempt one, runs 002 through 014: \$29.22, 455 turns. Run 001, the interactive research fan-out, is extra and was recorded by hand.

Attempt two, runs 015 through 029: \$23.81, 536 turns, 3,281 seconds of agent time.

Total for both attempts, as recorded in `runs/cost-report.md` on 2026-09-18: \$53.03. Any stage run after that date adds its own row.

The receipts

Every call left a directory, `runs/NNN-<stage>/`, holding `prompt.md`, the exact prompt sent; `result.json`, the full headless output; `return.md`, what the agent said back; `exit-code`; and its own `cost-row.tsv`.

The cost report is regenerated from every `result.json`. That is how a pipeline is audited rather than trusted.

The one paper that says both

Kim and colleagues ran the same experiment both ways. Centralized multi-agent improved a decomposable task by about 80 percent, and every multi-agent variant made a sequential planning task 39 to 70 percent worse.

Kim et al. (2026), Towards a Science of Scaling Agent Systems, arXiv 2512.08296: +80.8% on decomposable financial reasoning with a centralized design; sequential planning degraded by 39 to 70% across all four multi-agent variants; 260 configurations, 6 benchmarks, 3 model families.

Task shape decides

Task shape decides. That is an inference from the result, not a recommendation the paper makes. Work that decomposes into independent pieces is a candidate for fan-out; work in which each step depends on the last belongs in one context window.

The evidence in favor

The case in favor comes from Anthropic's research system, a lead agent plus subagents, which did about 90 percent better than a single agent on breadth research, at about 15 times the tokens of a chat.

Anthropic (June 2025), How we built our multi-agent research system, anthropic.com/engineering: outperformed single-agent Claude Opus 4 by 90.2% on an internal breadth-research evaluation; multi-agent systems use about 15× more tokens than chats. A vendor claim, not independently replicated.

The evidence against: equal budgets

The case against comes from two papers. In the first, at an equal thinking budget, a single agent matched or beat five multi-agent designs, and multi-agent won only when most of the context had been masked or corrupted.

Tran and Kiela (April 2026), Single-Agent LLMs Outperform Multi-Agent Systems on Multi-Hop Reasoning Under Equal Thinking Token Budgets, arXiv 2604.02460: at a 5,000-token budget, single-agent scored 0.427 against 0.386 for sequential multi-agent, aggregated over three model families; multi-agent won only with up to 70% of the context masked.

The evidence against: cost

In the second, one agent framework cost over 50 times a simple baseline, and the baseline was more accurate.

Kapoor et al. (2024), AI Agents That Matter, arXiv 2407.01502: LATS cost over 50 times more than the paper's "Warming" baseline on HumanEval, and Warming scored 93.2% to LATS's 88.0%.

The honest summary is that the outcome depends on the task, and the token multiple is real either way.

How it fails

The MAST study looked at where multi-agent systems break: over 1,600 traces across seven frameworks, fourteen failure modes in three categories. The categories are system design at 44.2 percent, inter-agent misalignment at 32.3 percent, and task verification at 23.5 percent.

Most failures are specification and coordination problems, not model errors.

Cemri et al. (2025), Why Do Multi-Agent LLM Systems Fail? (MAST), arXiv 2503.13657.

How it fails in practice

First, subagents do not see the orchestrator's conversation. They must be given what they need in the prompt, and asked for a summary in return, not a dump.

Second, two agents editing one file; the shared cost log in attempt one is an instance.

Third, agents spawning agents instead of a script calling agents. Once the model decides the order, reproducibility is lost.

A simple example to run first

The example is one read-only subagent. Given ten target names, it queries the NASA Exoplanet Archive with astroquery and returns a table under a token cap. It has two tools, Read and Bash, and the cheapest model. The description is cut short with an ellipsis; the full file is in `examples/`.

```
---  
name: exoplanet-lookup  
description: Read-only lookup of confirmed exoplanet parameters from the  
    NASA Exoplanet Archive for a given list of planet names. [...]  
tools: Read, Bash  
model: haiku  
---
```

The example: the instructions

Below the frontmatter are three rules in plain English: read the file, query the archive, and say "not found" rather than invent a number.

You are given a path to a text file with one exoplanet name per line (up to ten names).

Do this:

1. Read the file.
2. For each name, query the NASA Exoplanet Archive (table `pscomppars`) with astroquery for columns: `pl\_name, hostname, pl\_orbper, pl\_rade, pl\_bmasse, disc\_year`. One Python process, one query per name (or a single `where` clause with all names), is fine.
3. If a name has no match, say "not found" in that row. Do not invent numbers.

The example: the return format

The return format follows. This is the part that keeps the subagent from dumping its whole session back into the orchestrator's window: one table, a source line, and a token cap.

```
Return only a single compact markdown table, one row per input name, columns:  
`name | host | period_days | radius_earth | mass_earth | disc_year`.  
Round numbers to 3 significant figures. No prose before or after the table, except a one-line  
header stating the source ("NASA Exoplanet Archive, pscomppars"). Keep the whole reply under  
600 tokens.
```

The example: the command

The call is one command, headless, with a one-dollar cap. The output is JSON, so the cost and the token counts come back with the answer.

```
env -u CLAUDECODE claude -p --agent exoplanet-lookup --allowedTools "Read,Bash" \  
  --max-budget-usd 1 --output-format json \  
  "Look up the exoplanets listed one per line in targets.txt in the NASA Exoplanet Archive and return the table."
```

The example: the real result

NASA Exoplanet Archive, pscomppars

name	host	period_days	radius_earth	mass_earth	disc_year
Kepler-10 b	Kepler-10	0.837	1.47	3.24	2011
TRAPPIST-1 e	TRAPPIST-1	6.10	0.920	0.692	2017
Proxima Cen b	Proxima Cen	11.2	1.02	1.05	2016

The reply was ten rows and about 250 tokens. The model was `claude-haiku-4-5`, the cost \$0.0751, the wall time 82.5 seconds over 8 turns. One caveat: total output including thinking was 7,321 tokens, so the cap applies to the table returned, not to everything the agent emits.

Cost, and when it is worth it

Multi-agent runs three to fifteen times a single chat's tokens, by Anthropic's own figures, from two of their posts.

Anthropic (June 2025), How we built our multi-agent research system, anthropic.com/engineering: multi-agent systems use about 15× more tokens than chats. Anthropic, Phillips et al. (January 2026), Building multi-agent systems: when and how to use them, claude.com/blog: multi-agent implementations typically use 3 to 10× more tokens. Vendor figures; the range on the slide spans both.

Spending the multiple

Cheap or local models belong on subagents, and the frontier model on the orchestrator. Claude Code's subagent frontmatter takes a `model` field, so a cheap model can be pinned on a subagent, for example a custom Explore pinned to Haiku, while the orchestrator stays on the frontier model. Model selection and pricing are covered elsewhere in the workshop and are not repeated here.

Anthropic (2026), Create custom subagents, Claude Code documentation: `model` takes haiku, sonnet, opus, fable, or inherit; the built-in Explore inherits the main conversation's model by default unless a custom Explore is defined and pinned to a cheaper one (fetched 2026-09-17).

The difficult example

The difficult example is this pipeline: ten agents, a script that owns order, loop, and isolation, and a log for every call.

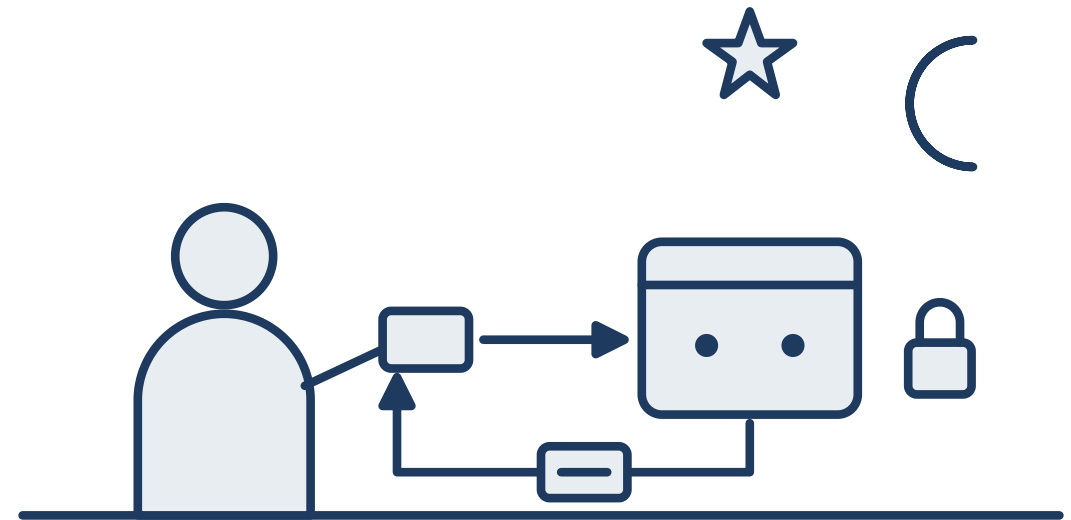
Multi-agent is worth it when the work exceeds one window, the pieces are independent, the output needs an independent check, and the token multiple is affordable. Otherwise, one agent suffices. Most of my own work is still one agent.

Close

More context than intelligence. Most tasks need one agent.

A first step is to pull one bounded, read-only task into its own agent file, run it once headless with a budget cap, and check that the return is short.

One closing line: many unsupervised agents require sandboxing, which is the subject of the following session.



Sources: segments A and B

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Anthropic (2026), Models overview, Claude Platform docs: <https://platform.claude.com/docs/en/about-claude/models>

Liu et al. (2023), Lost in the Middle: <https://arxiv.org/abs/2307.03172>

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